DistRL Market Maker — Training & Evaluation Results

5 RL agents vs. 3 analytical baselines across 3 volatility regimes in an ABIDES-Gym limit-order-book simulator, plus out-of-distribution transfer, CVaR risk-sensitivity, and state-representation ablations.

`seed=42` `budget=500 ep` `encoder=handcrafted` `reward=asymmetric`

Held-out performance · Training/eval curves · Training loss · Baseline replicates · AS-recovery · OOD transfer · Simulator validation · CVaR efficient frontier · Encoder ablation · Recurrent vs. snapshot · Reward comparison · Full results table

RL RUNS
15
5 agents × 3 regimes

EARLY STOPPED
8 / 15
775 episodes saved (10%)

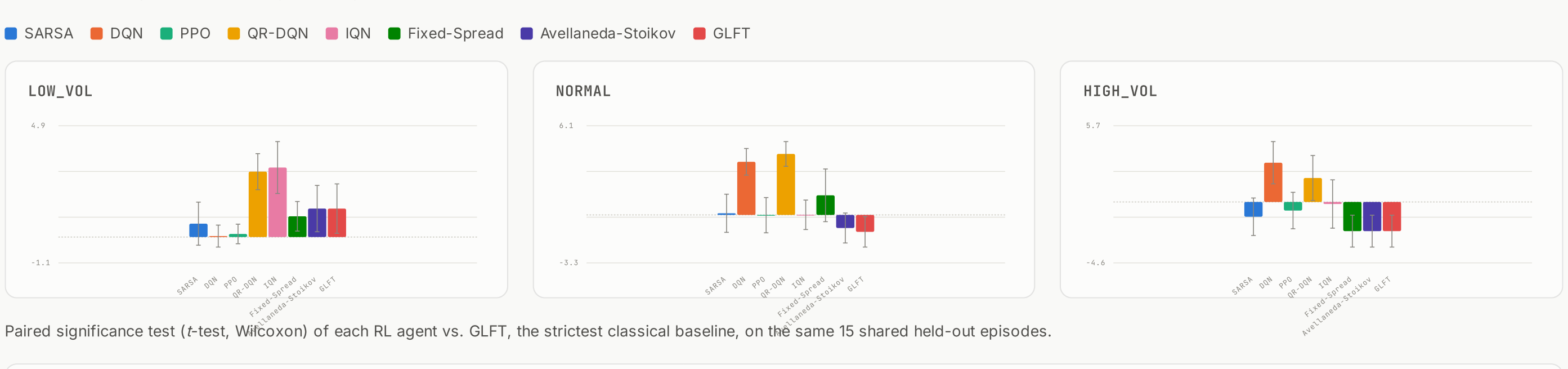
BEST HELD-OUT SHARPE (RL)
+4.194
QR-DQN · normal · 15 unseen episodes

BEST HELD-OUT SHARPE (BASELINE)
+1.348
Fixed-Spread · normal

REGIMES WHERE AN RL AGENT BEATS GLFT
3 / 3
on held-out data, GLFT is the strictest reference

Held-out test performance — the real comparison

Each model's best checkpoint (or fixed policy) evaluated on 15 held-out episodes never touched by training or selection, with every model seeing the identical market conditions per regime. This is the headline result — everything below is a training-time diagnostic.

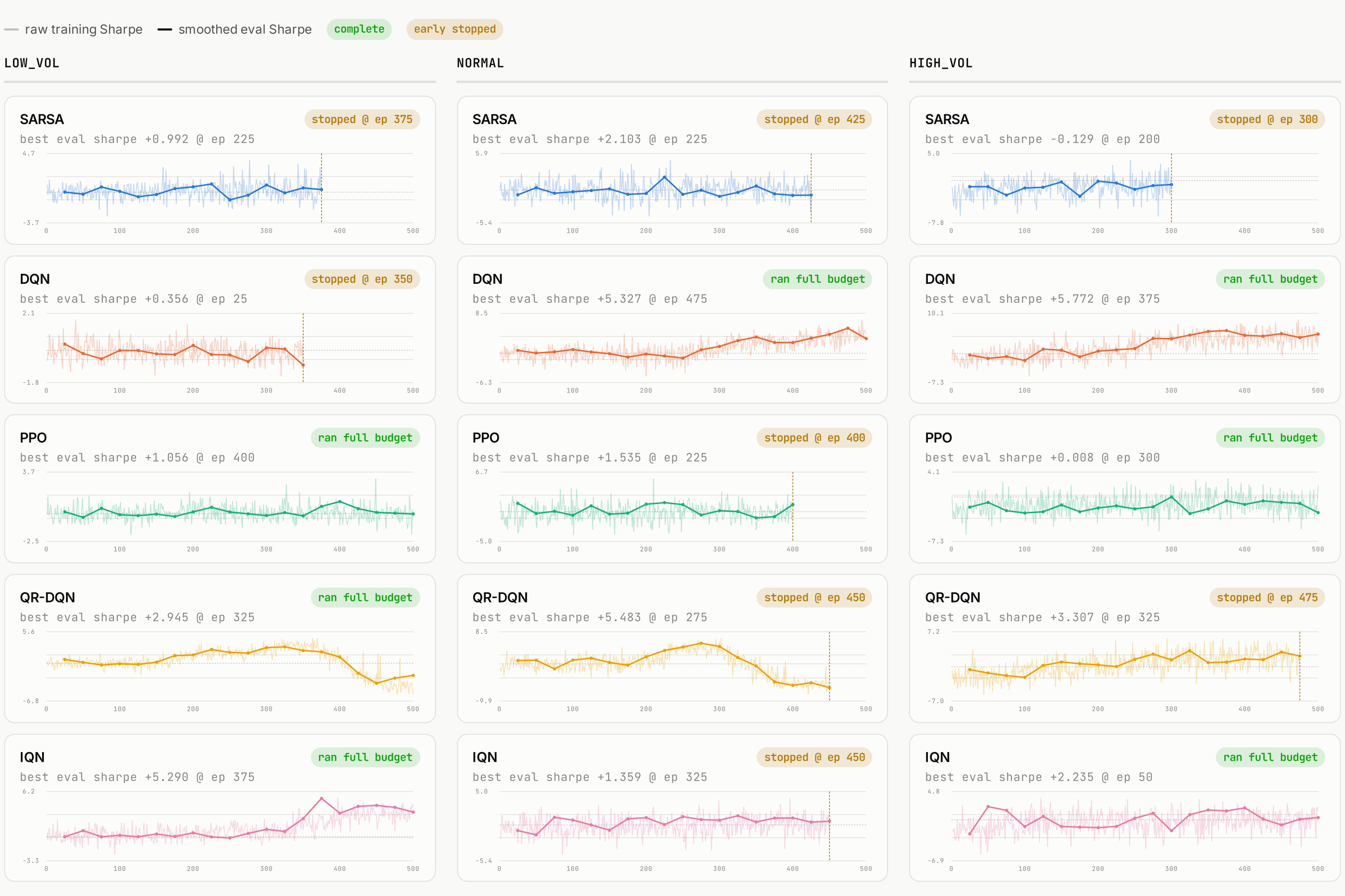


Paired significance test (*t*-test, Wilcoxon) of each RL agent vs. GLFT, the strictest classical baseline, on the same 15 shared held-out episodes.

AGENT	REGIME	MEAN Δ VS. GLFT	*T*-TEST P	WILCOXON P	SIGNIFICANTLY BEATS GLFT?
SARSA	high_vol	+1.075	0.0153	0.0256	yes, p<0.05
DQN	high_vol	+5.112	0.0000	0.0001	yes, p<0.05
PPO	high_vol	+1.538	0.0021	0.0034	yes, p<0.05
QR-DQN	high_vol	+3.971	0.0000	0.0001	yes, p<0.05
IQN	high_vol	+2.028	0.0014	0.0026	yes, p<0.05
SARSA	low_vol	-0.657	0.0799	0.0946	no
DQN	low_vol	-1.199	0.0029	0.0034	no
PPO	low_vol	-1.106	0.0006	0.0009	no
QR-DQN	low_vol	+1.613	0.0000	0.0001	yes, p<0.05
IQN	low_vol	+1.792	0.0001	0.0001	yes, p<0.05
SARSA	normal	+1.278	0.0003	0.0001	yes, p<0.05
DQN	normal	+4.818	0.0000	0.0001	yes, p<0.05
PPO	normal	+1.146	0.0162	0.0181	yes, p<0.05
QR-DQN	normal	+5.358	0.0000	0.0001	yes, p<0.05
IQN	normal	+1.173	0.0025	0.0009	yes, p<0.05

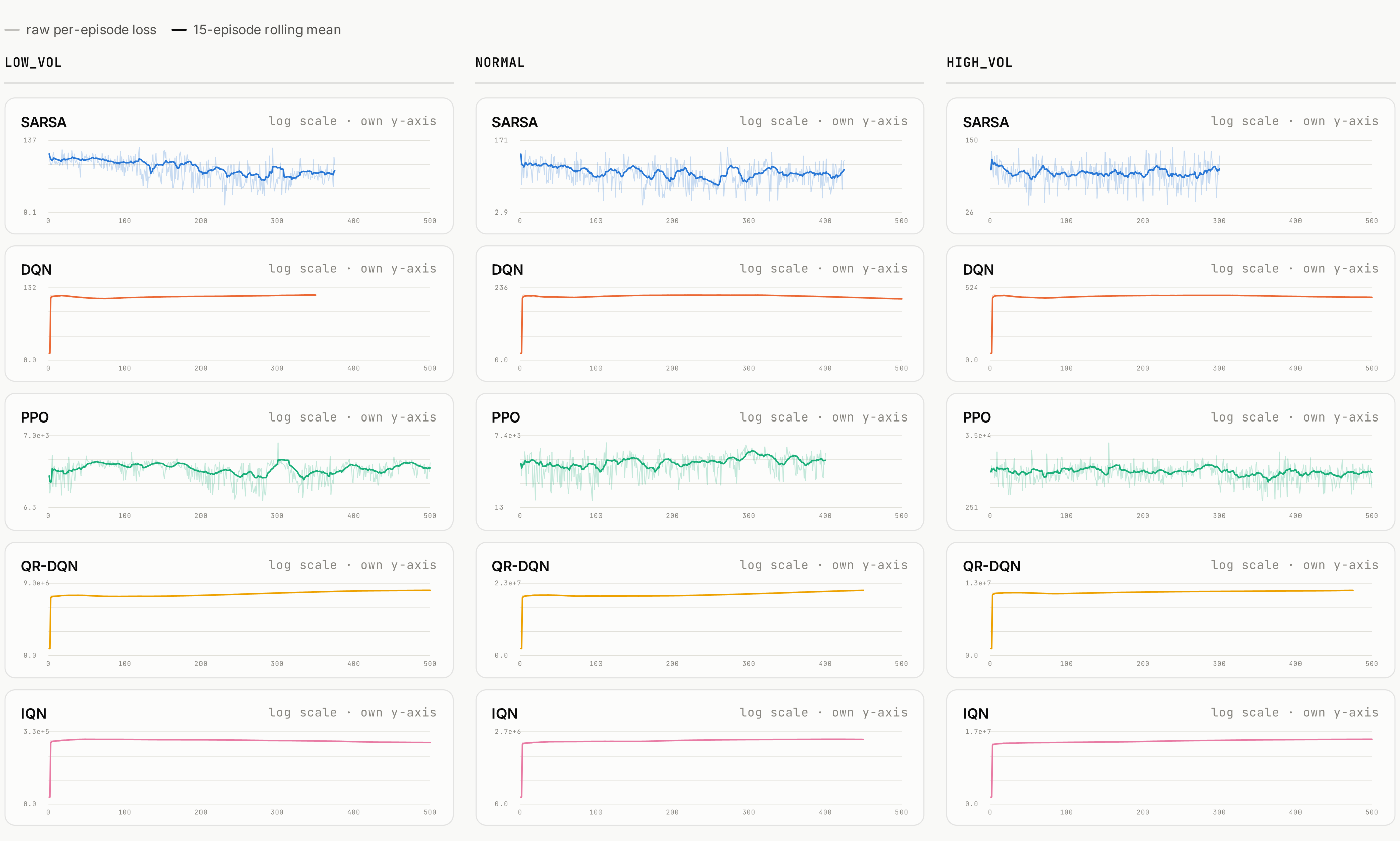
Training & evaluation curves — RL agents

Raw per-episode training Sharpe vs. smoothed eval Sharpe, the signal early stopping watches, for each RL agent per regime.



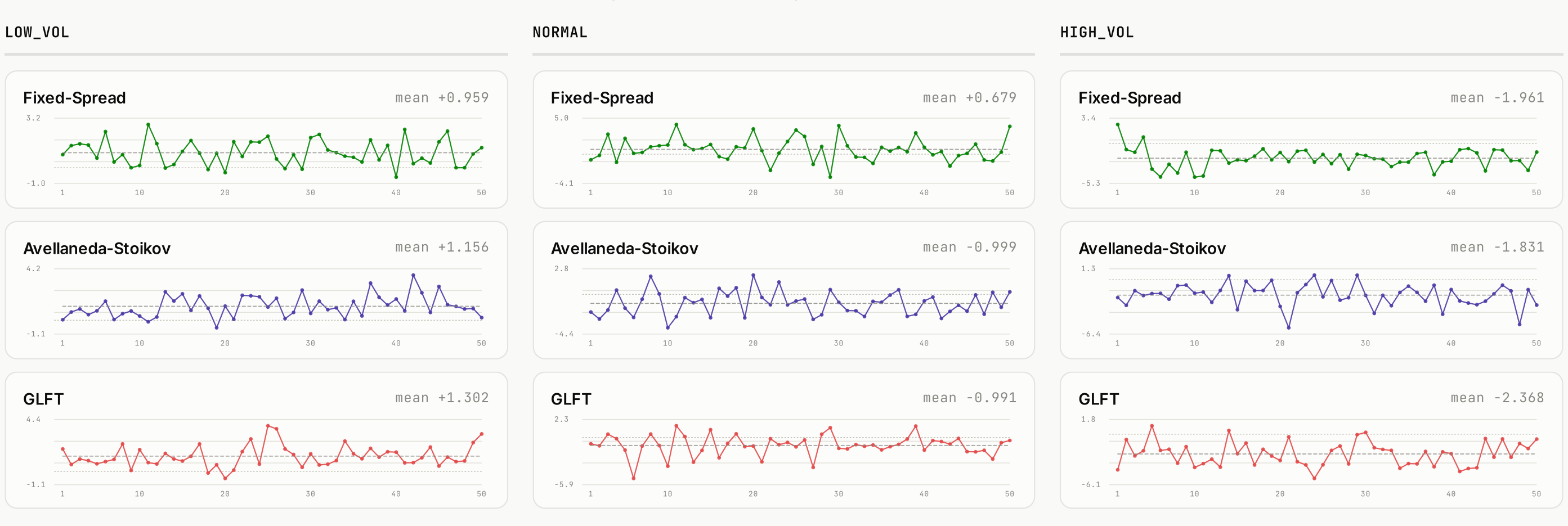
Training loss / objective — RL agents

Each agent's own training-objective loss on a log scale — not comparable across algorithms, since each optimizes a different loss function.



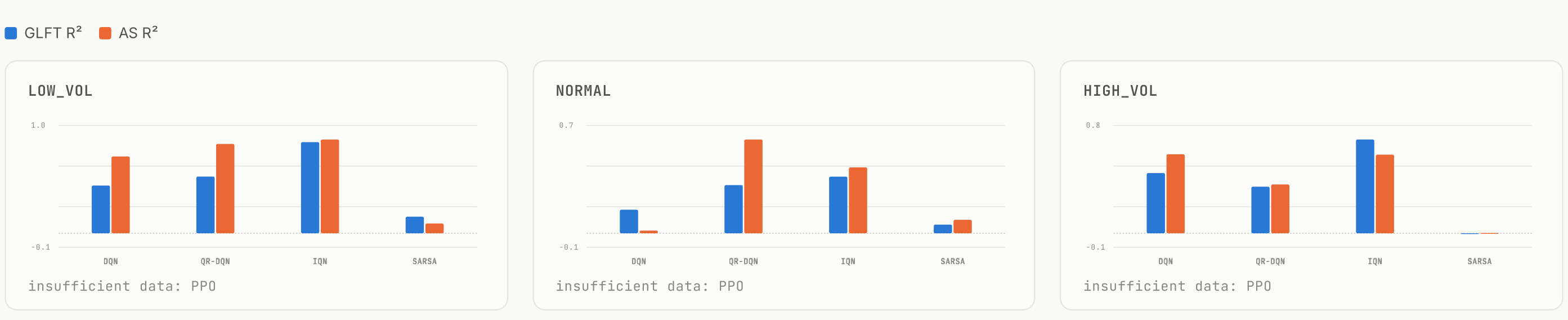
Baseline replicate performance

Performance variance across 50 independent replicates of each fixed (non-learning) baseline policy, per regime.



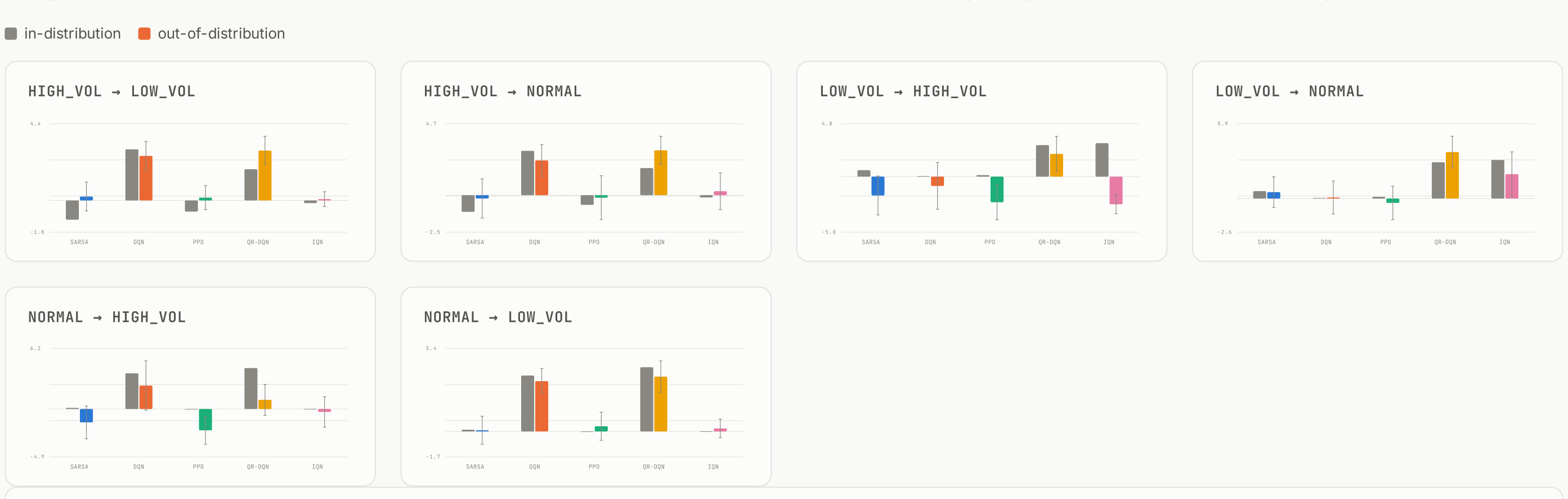
AS-recovery: does the agent rediscover analytical inventory-skew quoting?

How closely each agent's inventory-skew quoting matches the closed-form AS/GLFT curves (R^2) — a sanity check on learned behavior, not a target to beat.



OOD transfer: agents evaluated outside their training regime

Each agent's checkpoint from one regime dropped into a different regime with no fine-tuning, across all 6 directed pairs among low_vol/normal/high_vol — tests generalization to market conditions never seen in training. QR-DQN stays profitable in all 6 of 6 transfers and actually improves on its in-distribution Sharpe in 3 of them; transferring *into* high_vol is the hardest direction for most agents.



SCENARIO	AGENT	IN-DIST HELD-OUT SHARPE	OOD HELD-OUT SHARPE	ABSOLUTE Δ	DEGRADATION % (UNSTABLE NEAR ZERO)
high_vol → low_vol	DQN	+2.933	+2.559	-0.374	+13%
high_vol → low_vol	IQN	-0.151	+0.076	+0.227	-150%
high_vol → low_vol	PPO	-0.641	+0.163	+0.804	-125%
high_vol → low_vol	QR-DQN	+1.793	+2.862	+1.069	-60%
high_vol → low_vol	SARSA	-1.104	+0.224	+1.328	-120%
high_vol → normal	DQN	+2.933	+2.300	-0.633	+22%
high_vol → normal	IQN	-0.151	+0.262	+0.413	-273%
high_vol → normal	PPO	-0.641	-0.167	+0.474	-74%
high_vol → normal	SARSA	-1.104	-0.220	+0.884	-80%
high_vol → normal	QR-DQN	+1.793	+2.970	+1.178	-66%
low_vol → high_vol	IQN	+3.032	-2.505	-5.537	+183%
low_vol → high_vol	PPO	+0.134	-2.320	-2.454	+1832%
low_vol → high_vol	SARSA	+0.583	-1.710	-2.293	+393%
low_vol → high_vol	DQN	+0.041	-0.845	-0.886	n/a (in-dist ≈ 0)
low_vol → high_vol	QR-DQN	+2.853	+2.059	-0.794	+28%
low_vol → normal	IQN	+3.032	+1.913	-1.119	+37%
low_vol → normal	PPO	+0.134	-0.341	-0.475	+355%
low_vol → normal	SARSA	+0.583	+0.502	-0.081	+14%
low_vol → normal	DQN	+0.041	+0.082	+0.041	n/a (in-dist ≈ 0)
low_vol → normal	QR-DQN	+2.853	+3.651	+0.798	-28%
normal → high_vol	QR-DQN	+4.194	+0.939	-3.255	+78%
normal → high_vol	PPO	-0.018	-2.184	-2.166	n/a (in-dist ≈ 0)
normal → high_vol	SARSA	+0.114	-1.370	-1.484	+1307%
normal → high_vol	DQN	+3.653	+2.400	-1.253	+34%
normal → high_vol	IQN	+0.008	-0.298	-0.306	n/a (in-dist ≈ 0)
normal → low_vol	QR-DQN	+4.194	+3.581	-0.613	+15%
normal → low_vol	DQN	+3.653	+3.286	-0.367	+10%
normal → low_vol	SARSA	+0.114	+0.075	-0.039	+34%
normal → low_vol	IQN	+0.008	+0.197	+0.189	n/a (in-dist ≈ 0)
normal → low_vol	PPO	-0.018	+0.338	+0.356	n/a (in-dist ≈ 0)

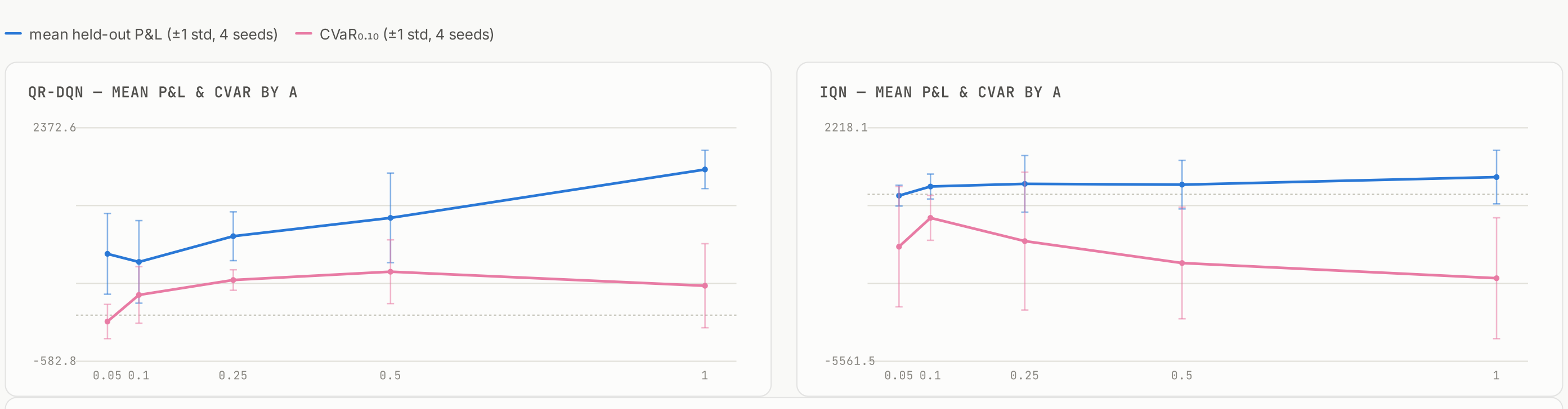
Simulator validation — stylized facts

A random-policy audit of the simulator itself against 5 standard empirical market-microstructure regularities, independent of any trained agent — validates that the environment produces realistic dynamics to learn from.

CHECK	LOW_VOL	NORMAL	HIGH_VOL
Fat-tailed returns	pass	pass	pass
Volatility clustering	pass	pass	pass
Spread autocorrelation	fail	pass	pass
Price impact	pass	pass	pass
Queue-imbalance predictability	pass	pass	pass
Overall	PASS	PASS	PASS

CVaR efficient frontier — QR-DQN & IQN, normal regime

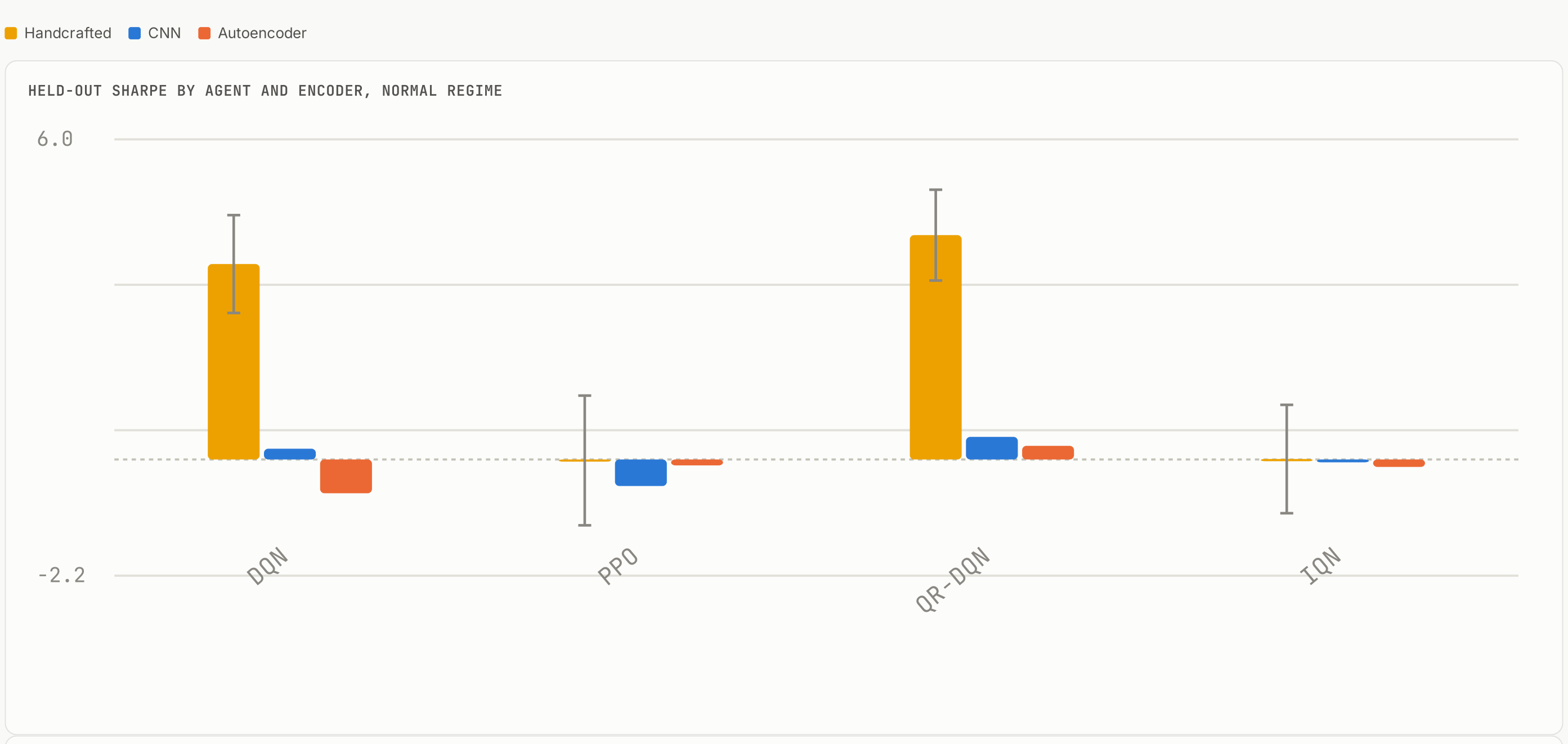
Mean P&L vs. tail risk ($\text{CVaR}_{0.10}$) across the CVaR-α sweep, averaged over 4 training seeds per setting. Seed variance is large relative to the gaps between adjacent α values, and CVaR-α doesn't produce a clean risk/return tradeoff at this training budget: for IQN the risk-neutral limit (α=1.0) has the highest mean P&L and the worst tail, but for QR-DQN mean P&L rises with α while the tail is best at interior values, so more conservative settings do not improve it.



AGENT	A	MEAN P&L	CVAR$_{0.10}$	MEAN SHARPE	SEEDS
QR-DQN	0.05	+776.715 ± +511.308	-79.873 ± +216.926	+2.086 ± +0.877	4
QR-DQN	0.10	+674.768 ± +522.657	+256.559 ± +357.865	+2.890 ± +1.361	4
QR-DQN	0.25	+999.651 ± +309.139	+445.192 ± +130.058	+3.893 ± +0.266	4
QR-DQN	0.50	+1231.978 ± +566.948	+550.346 ± +403.533	+3.692 ± +0.340	4
QR-DQN	1.00	+1844.705 ± +241.930	+372.841 ± +532.208	+3.024 ± +0.836	4
IQN	0.05	-44.975 ± +346.993	-1745.284 ± +2002.888	+0.131 ± +0.286	4
IQN	0.10	+259.522 ± +415.394	-784.023 ± +748.024	+0.557 ± +0.892	4
IQN	0.25	+347.859 ± +942.426	-1559.062 ± +2296.564	+1.927 ± +2.604	4
IQN	0.50	+322.809 ± +809.892	-2289.486 ± +1858.199	+0.565 ± +1.219	4
IQN	1.00	+573.933 ± +891.274	-2795.150 ± +2013.515	+0.476 ± +0.848	4

Encoder ablation — handcrafted vs. CNN vs. autoencoder, normal regime

Handcrafted features vs. two learned snapshot representations (CNN, autoencoder), for all 4 neural agents. Handcrafted significantly wins for QR-DQN and DQN (p<0.001), with no significant difference for PPO or IQN — because those two agents' own handcrafted baselines are themselves weak, not because their learned encoders work better.

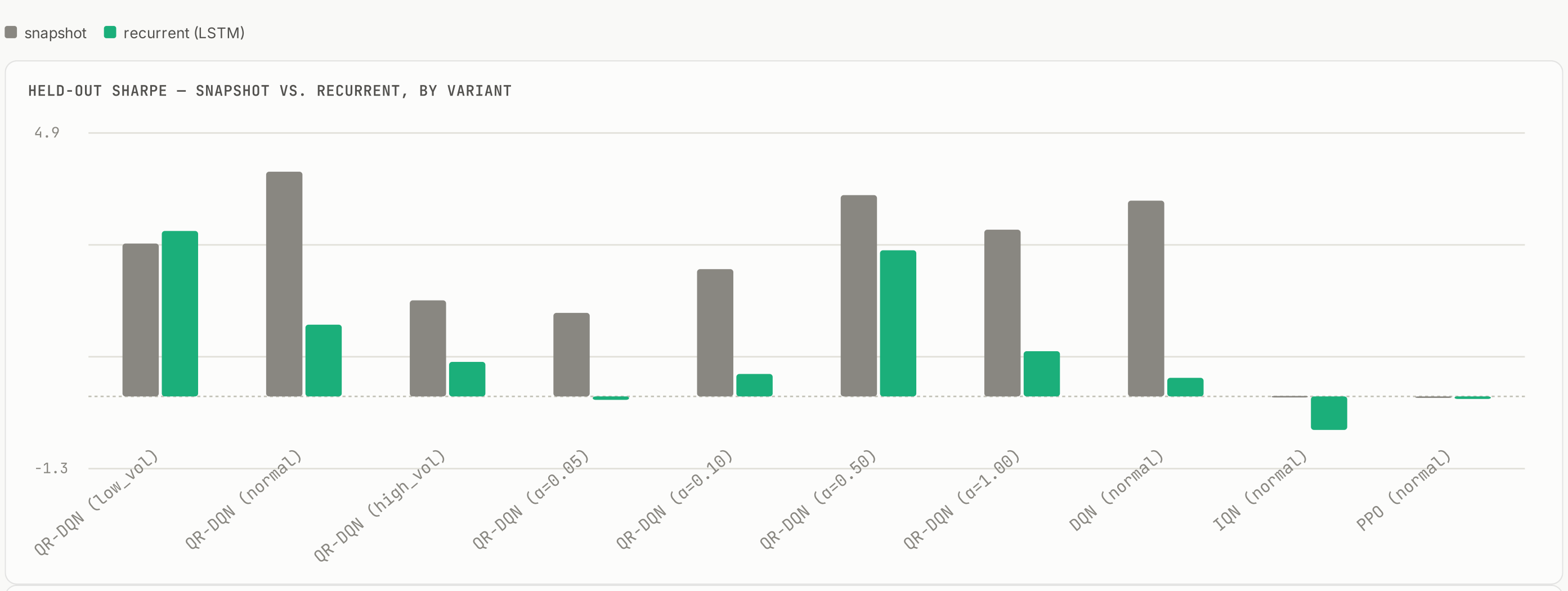


AGENT	ENCODER	ENCODER SHARPE	HANDCRAFTED SHARPE	MEAN Δ	*T*-TEST P	WILCOXON P	SIGNIFICANTLY DIFFERENT?
DQN	CNN	+0.201	+3.653	-3.452	0.000002	0.000061	yes, p<0.05
DQN	Autoencoder	-0.631	+3.653	-4.284	0.000000	0.000061	yes, p<0.05
PPO	CNN	-0.496	-0.018	-0.478	0.219308	0.207764	no
PPO	Autoencoder	-0.110	-0.018	-0.092	0.793469	1.000000	no
QR-DQN	CNN	+0.422	+4.194	-3.772	0.000003	0.000122	yes, p<0.05
QR-DQN	Autoencoder	+0.255	+4.194	-3.939	0.000000	0.000061	yes, p<0.05
IQN	CNN	-0.053	+0.008	-0.061	0.735782	0.977966	no
IQN	Autoencoder	-0.139	+0.008	-0.147	0.420485	0.454285	no

AGENT	ENCODER	HANDCRAFTED GLFT R²	ENCODER GLFT R²	Δ
DQN	CNN	0.158	0.542	+0.384
DQN	Autoencoder	0.158	0.739	+0.580
PPO	CNN	—	0.018	—
PPO	Autoencoder	—	—	—
QR-DQN	CNN	0.324	0.095	-0.228
QR-DQN	Autoencoder	0.324	0.488	+0.165
IQN	CNN	0.380	0.088	-0.292
IQN	Autoencoder	0.380	0.004	-0.376

Recurrent vs. snapshot: does temporal memory help?

Each agent's standard snapshot-encoder policy vs. the same agent with an LSTM backbone in its place, across 10 agent/regime/CVaR-risk combinations. Recurrent integration never significantly helps and significantly hurts seven of the ten: QR-DQN, DQN, and IQN in the `normal` regime, and QR-DQN at every CVaR α tried (0.05, 0.10, 0.50, 1.0). QR-DQN in `low_vol`/`high_vol` and PPO show no significant difference.

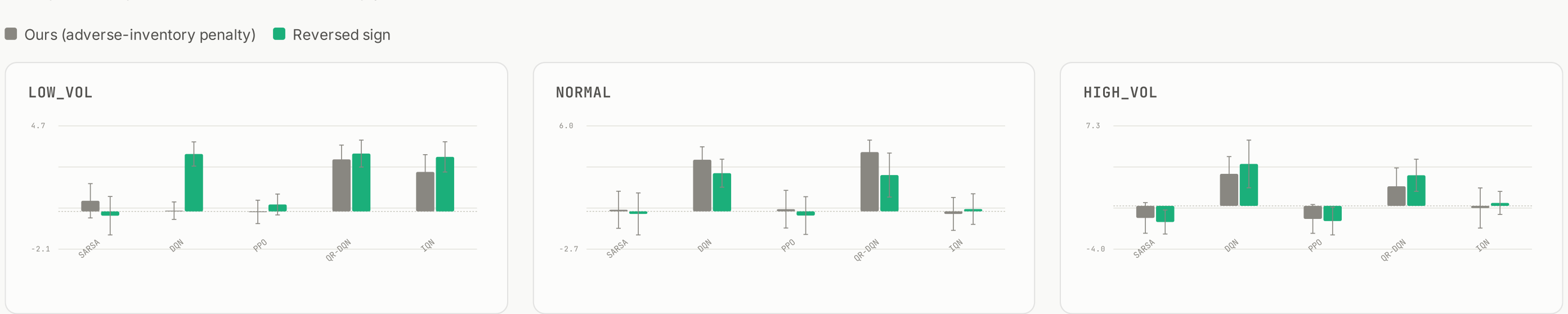


VARIANT	SNAPSHOT SHARPE	RECURRENT SHARPE	MEAN Δ	*T*-TEST P	WILCOXON P	SIGNIFICANTLY DIFFERENT?
QR-DQN (low_vol)	+2.853	+3.087	+0.234	0.441011	0.561401	no
QR-DQN (normal)	+4.194	+1.339	-2.855	0.000077	0.000122	yes, p<0.05
QR-DQN (high_vol)	+1.793	+0.644	-1.149	0.089223	0.276855	no
QR-DQN (α=0.05)	+1.559	-0.064	-1.623	0.014735	0.018066	yes, p<0.05
QR-DQN (α=0.10)	+2.375	+0.419	-1.956	0.005976	0.005371	yes, p<0.05
QR-DQN (α=0.50)	+3.756	+2.727	-1.029	0.032221	0.025574	yes, p<0.05
QR-DQN (α=1.00)	+3.112	+0.843	-2.268	0.000179	0.000122	yes, p<0.05
DQN (normal)	+3.653	+0.346	-3.307	0.000000	0.000061	yes, p<0.05
IQN (normal)	+0.008	-0.626	-0.634	0.040146	0.041260	yes, p<0.05
PPO (normal)	-0.018	-0.047	-0.029	0.955714	0.803955	no

VARIANT	SNAPSHOT GLFT R²	RECURRENT GLFT R²	Δ
QR-DQN (low_vol)	—	0.846	—
QR-DQN (normal)	—	0.024	—
QR-DQN (high_vol)	—	0.173	—
QR-DQN (α=0.05)	—	0.574	—
QR-DQN (α=0.10)	—	0.717	—
QR-DQN (α=0.50)	—	0.378	—
QR-DQN (α=1.00)	—	0.011	—
DQN (normal)	—	0.452	—
IQN (normal)	—	0.011	—

Reward comparison: our adverse-inventory penalty vs. the reversed sign

Our reward penalises inventory losses and keeps gains; the comparator reverses that sign, keeping losses and penalising gains. Across 15 agent×regime pairs (seed 42), ours wins significantly in 4, the reversed sign wins significantly in 4, and the rest show no significant difference — but QR-DQN significantly beats GLFT in the `normal` regime under either sign. Three extra seeds for QR-DQN and IQN in `normal` show the same pair can flip from seed to seed, most sharply for IQN.



Paired significance test (seed 42), each reward vs. GLFT and against each other on the same 15 held-out episodes.

AGENT	REGIME	OURS SHARPE	REVERSED SHARPE	MEAN Δ (OURS−REV)	PAIR WILCOXON P	SIG. DIFFERENT?	OURS VS GLFT	REVERSED VS GLFT
SARSA	low_vol	+0.583	-0.235	+0.818	0.055359	no	-0.657	-1.475
DQN	low_vol	+0.041	+3.148	-3.107	0.000061	yes, p<0.05	-1.199	+1.908
PPO	low_vol	-0.026	+0.381	-0.407	0.135376	no	-1.266	-0.859
QR-DQN	low_vol	+2.853	+3.173	-0.320	0.168823	no	+1.613	+1.933
IQN	low_vol	+2.166	+2.990	-0.824	0.018066	yes, p<0.05	+0.926	+1.750
SARSA	normal	+0.114	-0.181	+0.295	0.599487	no	+1.278	+0.983
DQN	normal	+3.653	+2.702	+0.952	0.018066	yes, p<0.05	+4.818	+3.866
PPO	normal	+0.152	-0.297	+0.449	0.599487	no	+1.317	+0.867
QR-DQN	normal	+4.194	+2.568	+1.626	0.001526	yes, p<0.05	+5.358	+3.732
IQN	normal	-0.182	+0.162	-0.344	0.072998	no	+0.982	+1.326
SARSA	high_vol	-1.104	-1.477	+0.373	0.276855	no	+1.075	+0.702
DQN	high_vol	+2.933	+3.840	-0.907	0.008362	yes, p<0.05	+5.112	+6.019
PPO	high_vol	-1.193	-1.388	+0.195	0.524475	no	+0.986	+0.790
QR-DQN	high_vol	+1.793	+2.804	-1.011	0.030151	yes, p<0.05	+3.971	+4.983
IQN	high_vol	-0.195	+0.269	-0.465	0.638672	no	+1.984	+2.448

Three extra seeds for QR-DQN and IQN in the `normal` regime — the same head-to-head, repeated to check it isn't a seed-42 artifact.

AGENT	SEED	OURS SHARPE	REVERSED SHARPE	MEAN Δ	PAIR WILCOXON P	SIG. DIFFERENT?
QR-DQN	1042	+3.779	+4.026	-0.246	0.359131	no
QR-DQN	2042	+3.586	+3.667	-0.081	0.890381	no
QR-DQN	3042	+4.015	+4.360	-0.345	0.524475	no
IQN	1042	+4.363	-0.349	+4.712	0.000061	yes, p<0.05
IQN	2042	+0.039	+0.033	+0.006	0.761536	no
IQN	3042	+3.518	-0.389	+3.907	0.000061	yes, p<0.05

Full results table

Every figure charted above, in one place.

AGENT	TYPE	REGIME	EPISODES	STATUS	HELD-OUT SHARPE (15 UNSEEN EP, FAIR)	± STD	PEAK EVAL SHARPE	@ EP	FINAL EVAL SHARPE (LAST 3, IN-SAMPLE)	TRAIN-HIST. MEAN (DIAGNOSTIC ONLY)	GLFT R²	AS R²
SARSA	RL	low_vol	375 / 500	early stopped	**+0.583**	0.94	+0.992	225	+0.248	+0.149	0.146	0.086
SARSA	RL	normal	425 / 500	early stopped	**+0.114**	1.31	+2.103	225	-0.777	+0.044	0.058	0.090
SARSA	RL	high_vol	300 / 500	early stopped	**-1.104**	1.40	-0.129	200	-1.100	-1.346	0.000	0.002
DQN	RL	low_vol	350 / 500	early stopped	**+0.041**	0.48	+0.356	25	-0.209	-0.080	0.421	0.677
DQN	RL	normal	500 / 500	complete	**+3.653**	0.91	+5.327	475	+4.147	+0.993	0.158	0.018
DQN	RL	high_vol	500 / 500	complete	**+2.933**	1.58	+5.772	375	+4.633	+1.751	0.449	0.589
PPO	RL	low_vol	500 / 500	complete	**+0.134**	0.43	+1.056	400	+0.030	+0.038	—	—
PPO	RL	normal	400 / 500	early stopped	**-0.018**	1.21	+1.535	225	-0.229	-0.022	—	—
PPO	RL	high_vol	500 / 500	complete	**-0.641**	1.36	+0.008	300	-1.497	-1.275	—	—
QR-DQN	RL	low_vol	500 / 500	complete	**+2.853**	0.79	+2.945	325	-2.810	+1.124	0.500	0.788
QR-DQN	RL	normal	450 / 500	early stopped	**+4.194**	0.85	+5.483	275	-5.703	+2.403	0.324	0.630
QR-DQN	RL	high_vol	475 / 500	early stopped	**+1.793**	1.68	+3.307	325	+2.224	+0.328	0.347	0.364
IQN	RL	low_vol	500 / 500	complete	**+3.032**	1.13	+5.290	375	+3.921	+1.169	0.804	0.827
IQN	RL	normal	450 / 500	early stopped	**+0.008**	1.01	+1.359	325	+0.670	+2.006	0.380	0.443
IQN	RL	high_vol	500 / 500	complete	**-0.151**	1.81	+2.235	50	-0.128	-0.495	0.699	0.586
Fixed-Spread	baseline	low_vol	50 / 50	n/a (i.i.d.)	**+0.904**	0.65	—	—	—	+0.959	—	—
Fixed-Spread	baseline	normal	50 / 50	n/a (i.i.d.)	**+1.348**	1.81	—	—	—	+0.679	—	—
Fixed-Spread	baseline	high_vol	50 / 50	n/a (i.i.d.)	**-2.179**	1.19	—	—	—	-1.961	—	—
Avellaneda-Stoikov	baseline	low_vol	50 / 50	n/a (i.i.d.)	**+1.241**	1.01	—	—	—	+1.156	—	—
Avellaneda-Stoikov	baseline	normal	50 / 50	n/a (i.i.d.)	**-0.901**	1.03	—	—	—	-0.999	—	—
Avellaneda-Stoikov	baseline	high_vol	50 / 50	n/a (i.i.d.)	**-2.179**	1.19	—	—	—	-1.831	—	—
GLFT	baseline	low_vol	50 / 50	n/a (i.i.d.)	**+1.240**	1.08	—	—	—	+1.302	—	—
GLFT	baseline	normal	50 / 50	n/a (i.i.d.)	**-1.165**	1.05	—	—	—	-0.992	—	—
GLFT	baseline	high_vol	50 / 50	n/a (i.i.d.)	**-2.179**	1.19	—	—	—	-2.368	—	—

Scaled-down local sweep · ABIDES-Gym rmsc04 background · generated from `logs/` and `results/`.